

Intel 955X utilises a new standard in its memory controller known as MPT or Memory Pipeline Technology, which uses each memory channel optimally—resulting in a bandwidth higher than that of the 945.

Both chipset makers have come up with their own versions of RAID implementations in the Southbridge. Intel has what it calls Matrix RAID, which can combine RAID 0 and 1 arrays using only two drives. nVidia has nVraid, which, amazingly, is capable of spanning arrays across both ATA and SATA drives. This means, for example, that you can have partitions that span two drives, one of them SATA and the other ATA.

nVidia's hardware-accelerated Gigabit Ethernet controller and firewall are unchallenged by Intel ICH7R, because the latter lacks a network controller.

Support for high-definition sampling rates (24-bit at 196 Kbps) audio is the bone of contention for Intel on the audio front, which nVidia's AC'97 solution is incapable of providing. But neither of these support hardware-accelerated 3D audio.

Features

Dual-core processor support was available in the boards that sported the Intel 945G and 955X and the nVidia nForce4 SLI Intel Edition



MSI P4N Diamond

chipsets. Even if you have only a regular (single-core) processor, a board that supports dual-core should be preferred, keeping upgradeability in mind.

One thing to note here is that nForce4 SLI-based motherboards still do not enjoy full compatibility with dual-core processors from

? Jargon Buster

The BIOS: The Basic Input-Output System is a program located on the CMOS chip on the motherboard. The BIOS controls the primary functions of the hardware and facilitates low-level communication between the OS and the hardware. The BIOS can be upgraded to provide additional functionality.

Bluetooth: A wireless radio standard, primarily developed for lower power consumption. It has a short range of 10 m, 100 m or 400 m, depending on the power class. It has certain advantages over prevailing radio standards: Bluetooth-enabled devices can communicate with each other even if they are of different types—for example, MP3 players, mobile phones, laptops, headsets, printers, and more.

DDR RAM: Double Data Rate SD RAM is a type of memory that handles data at twice the speed of the older SD RAM. DDR SD RAM typically operates at 266 MHz, 333 MHz or 400 MHz effective speeds, while the actual speeds are 133, 166 and 200 MHz respectively.

DDR2 RAM: Double Data Rate 2 RAM is a memory type based on chips designed to run at speeds higher than that of DDR RAM, such as 533MHz and 667 MHz. Few motherboards support DDR2 RAM right now.

Dual-Channel: When two RAM modules are plugged into two slots marked for two different memory channels, then, theoretically, the data flows at twice the normal rate through the RAM. This is

known as a dual-channel memory configuration.

FSB: The Front Side Bus speed is the speed at which the different components of the computer speak to the CPU. The CPU's FSB is the most important, and changing it can dramatically affect the overall performance of the computer.

IDE: Integrated Drive Electronics is a standard interface for hard drives, optical drives and other physical storage devices. It is also known as ATA, which stands for Advanced Technology Attachment, and Parallel ATA. Various types are ATA33, ATA66, ATA100 and ATA133, based on transfer speeds. It uses a 40-pin connector cable that may consist of 40 or 80 conductors.

JBOD: Just a Bunch Of Disks is a disk configuration which, in the true sense, is not RAID at all. When two or more hard drives are in JBOD configuration, the entire space of all the disks can be combined into a single volume, or can be split into separate logical volumes. This does not provide fault tolerance, but there are certain advantages, such as no wastage of drive space and easier disaster recovery.

Matrix RAID: A technology developed by Intel, it achieves what RAID 0 and 1 achieve, using only two hard drives. It does so by creating two separate volumes in an array of two hard disks. It enables you to separate a two-disk array into two volumes, as if there were two separate sets of hard disks.

Multiplier: This is a number for a CPU that determines how fast the CPU operates. The CPU speed equals the FSB multiplied by the multiplier.

Northbridge: A chip on the motherboard that controls the FSB and memory. This is the largest chip on the motherboard and is normally covered by a heat sink, and sometimes actively cooled by a fan.

PCI: Peripheral Component Interconnect is a standard used on all current-generation motherboards and expansion cards used to connect the two devices. The data transfer rate of PCI bus can be up to 133 MBps.

PCI-Express (PCIe): Peripheral Component Interconnect Express is a new PCI standard that can transfer data at speeds equivalent to AGP 16x, which is roughly 4.3 GB/s.

RAID: Redundant Array of Independent Disks is a software or hardware configuration that employs two or more drives in combination for fault tolerance and performance. RAID 0, 1 and 0+1 are the commonly used types.

RAID 0: This involves only striping, and there is no data redundancy. It has the best performance but no fault tolerance. (Striping is a technique in which the drive's storage space is partitioned into units ranging from 512 bytes to several megabytes.) The stripes are interleaved and addressed in order.

RAID 1: This type of RAID implementation is also known as disk mirroring, which involves at least two